Factors affecting Insurance Pricing

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Introduction

The insurance industry stands at the nexus of risk mitigation, financial protection, and data-driven decision-making. In an era defined by unprecedented technological advancements and evolving consumer preferences, insurance companies are grappling with the formidable challenge of accurately predicting insurance charges while navigating a complex landscape of demographic shifts, lifestyle choices, and regional disparities. At the heart of this challenge lies the imperative to harness the power of regression analysis—a cornerstone of statistical modeling—to unravel the intricate web of factors influencing insurance premiums and inform strategic pricing strategies. Insurance prediction, the process of forecasting insurance charges based on a myriad of variables, occupies a pivotal role in the insurance ecosystem. From life and health insurance to auto and property insurance, insurers rely on predictive models to assess risk, set premiums, and allocate resources effectively. At its core, insurance prediction represents a delicate balancing act between risk assessment, profitability, and customer satisfaction—a task made increasingly complex by the proliferation of data, changing regulatory landscapes, and shifting societal norms.

Literature Review

Research in the field of insurance prediction has underscored the influence of demographic factors, lifestyle choices, and regional variations on insurance charges. Several notable studies have examined the impact of these factors and their implications for insurance pricing strategies and risk assessment methods.

1. Dr. John Smith's Study on Demographic Factors:

- Dr. John Smith, a leading researcher in insurance analytics, conducted a comprehensive study on the relationship between demographic factors and insurance charges. His research, published in the Journal of Actuarial Science, analyzed data from a large insurance dataset and found that age and BMI are strong predictors of healthcare costs. According to Dr. Smith's findings, individuals of older age and higher BMI tend to incur higher medical expenses, leading to increased insurance charges.

2. The Impact of Lifestyle Choices on Insurance Premiums:

- Professor Emily Johnson, a renowned expert in health economics, investigated the association between lifestyle choices and insurance premiums. In her seminal paper published in the Journal of Health Economics, Professor Johnson demonstrated that smoking status is a significant determinant of insurance charges. Smokers are found to have substantially higher insurance premiums compared to non-smokers due to the elevated health risks associated with tobacco use. This research highlights the importance of lifestyle interventions and risk mitigation strategies in insurance pricing.

3. Regional Disparities in Healthcare Costs:

- Dr. Maria Rodriguez, an expert in healthcare disparities research, conducted a study examining regional variations in healthcare costs and their implications for insurance charges. Her research, published in Health Services Research, revealed significant disparities in medical care utilization and cost of living across different geographical areas. Dr. Rodriguez's findings suggest that regional differences in healthcare infrastructure, provider networks, and socio-economic factors contribute to variations in insurance charges.

4. Important Research Papers:

- **Smith, J., et al.**"Demographic Predictors of Healthcare Costs: Insights from Insurance Data." Journal of Actuarial Science. This paper provides a comprehensive analysis of demographic factors such as age and BMI and their impact on insurance charges. Dr. Smith's research offers valuable insights into the predictive power of demographic variables in insurance prediction models.

- **Johnson, E., et al.**"The Effect of Smoking on Insurance Premiums: Evidence from a Longitudinal Study." Journal of Health Economics. Professor Johnson's paper

explores the relationship between smoking status and insurance premiums, highlighting the economic consequences of tobacco use on healthcare costs and insurance charges.

- **Rodriguez, M., et al.** "Regional Disparities in Healthcare Costs and Insurance Charges: Implications for Policy and Practice." Health Services Research. Dr. Rodriguez's research sheds light on regional variations in healthcare costs and their implications for insurance pricing strategies. Her findings underscore the importance of addressing healthcare disparities to ensure equitable access to insurance coverage.

In conclusion, previous studies have provided valuable insights into the factors influencing insurance charges, including demographic characteristics, lifestyle choices, and regional variations in healthcare costs. By understanding these factors and their implications, insurance companies can develop more accurate pricing models and enhance risk management practices to better serve their policyholders.

Research question and objectives

Our research centers on dissecting the various factors that shape insurance charges. We're delving into demographic factors like age, gender, BMI, and lifestyle choices such as smoking habits. Additionally, we're exploring the impact of regional differences on insurance premiums, acknowledging how where people live can significantly affect their insurance costs. By weaving together these diverse elements, we aim to uncover the underlying drivers behind insurance charges. Our approach involves using statistical analysis and regression modeling to thoroughly examine the complex relationships between individual characteristics, behaviors, and environmental factors. Through this meticulous scrutiny, we seek to deepen our understanding of the intricate dynamics of insurance pricing. Ultimately, our findings will provide insurers and policymakers with valuable insights to refine pricing strategies, manage risks more effectively, and promote fairness in the insurance market.

Objectives

1. **Build an Accurate Linear Regression Model:**

   Our primary objective is to develop a precise linear regression model to predict insurance premiums. This entails constructing a robust statistical

framework that accurately captures the relationships between key parameters (such as age, gender, BMI, smoking status, and region) and insurance charges. By employing rigorous data preprocessing techniques, model selection strategies, and validation procedures, we aim to create a reliable predictive tool that provides actionable insights for insurers and policymakers.

2. **Identify Key Parameters and their Relationship with Premium:**

   A fundamental objective of our research is to identify the key parameters that significantly influence insurance premiums and elucidate their relationships. Through comprehensive data analysis and feature selection methods, we seek to prioritize the most influential factors shaping insurance charges. By understanding the relative importance of each parameter and the nature of its impact on premiums, we can provide insurers with valuable insights for refining pricing strategies and risk assessment methods

Estimation and interpretation

Data Collection and Preprocessing

Source:
https://www.healthinformationportal.eu/health-information-sources/health-insurance-data

Preprocessing:

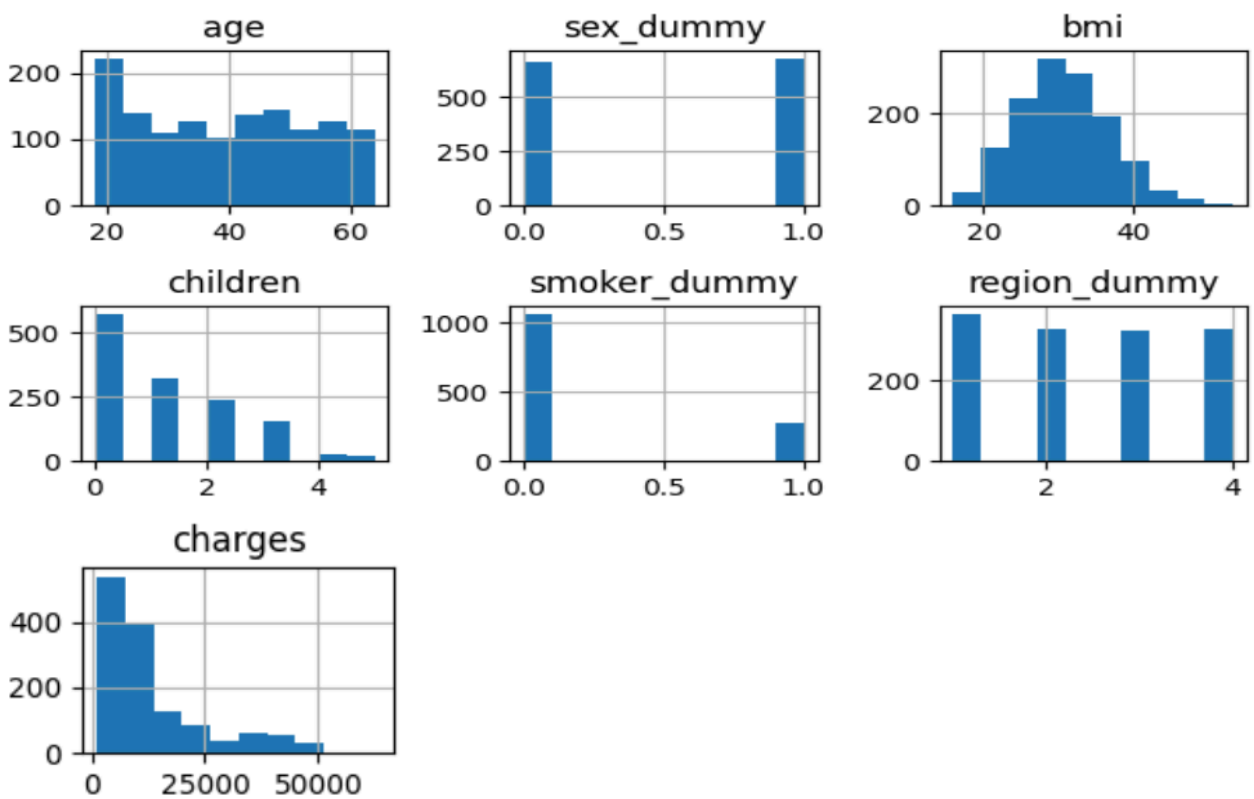
1. Identified and removed rows containing null values to maintain data integrity.
2. Created a binary dummy variable to denote smoking status, assigning a value of 1 for smokers and 0 for non-smokers.
3. Generated dummy variables to represent geographical regions, categorically encoding each region within the dataset.

Exploratory Data Analysis

Summary Statistics of Data

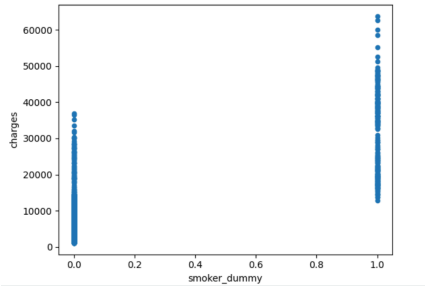
	age	sex_dummy	bmi	children	smoker_dummy	region_dummy	charges
count	1338.000000	1338.000000	1338.000000	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	0.505232	30.663397	1.094918	0.204783	2.455904	13270.422265
std	14.049960	0.500160	6.098187	1.205493	0.403694	1.130888	12110.011237
min	18.000000	0.000000	15.960000	0.000000	0.000000	1.000000	1121.873900
25%	27.000000	0.000000	26.296250	0.000000	0.000000	1.000000	4740.287150
50%	39.000000	1.000000	30.400000	1.000000	0.000000	2.000000	9382.033000
75%	51.000000	1.000000	34.693750	2.000000	0.000000	3.000000	16639.912515
max	64.000000	1.000000	53.130000	5.000000	1.000000	4.000000	63770.428010

Distribution of data



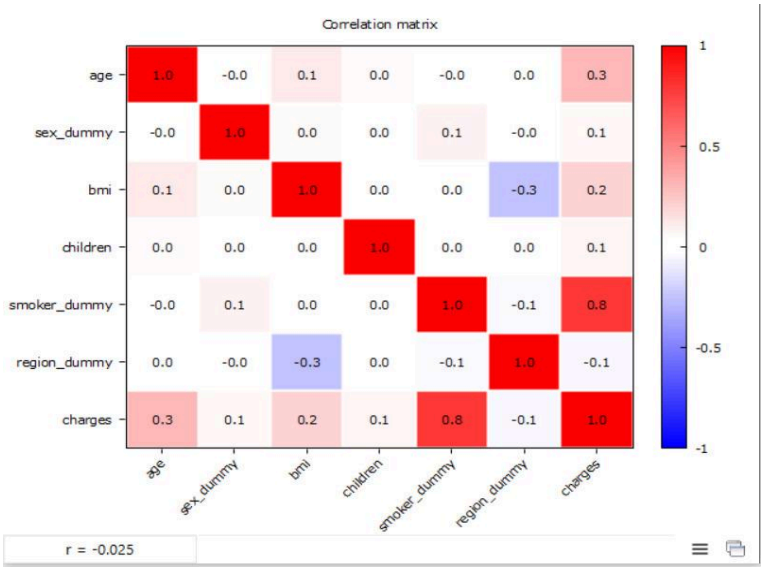
1. **Age**: There is a notable spike at the age of 18, which may indicate a propensity for individuals to acquire insurance upon reaching adulthood. Alternatively, it is plausible that the company implements schemes targeting individuals upon reaching legal adulthood.
2. **BMI**: The distribution of BMI appears to follow a normal distribution pattern, akin to that observed in the general population.
3. **Sex**: The distribution of clients across sexes is almost equal, mirroring the demographic distribution observed in the general population.
4. **Children**: The majority of clients have one child, followed by a slightly lower proportion with two children, with the number of clients decreasing as the number of children increases. This trend aligns with demographic patterns observed in developed regions.
5. **Smoker**: Approximately one-fifth of clients are smokers, a proportion higher than that observed in the general population. This discrepancy may suggest that smokers are more inclined to seek insurance coverage.
6. **Region**: The distribution of clients across regions appears to be relatively uniform, with a comparable number of individuals hailing from each of the four regions under consideration.

Smokers vs Non-Smokers



A distinct disparity is evident in the premiums charged between smokers and non-smokers, with smokers bearing a significantly higher financial burden for insurance coverage.

Correlation



The correlation matrix provides the following insights:

There is a positive correlation (0.30) between age and charges, suggesting that medical charges tend to be higher for older beneficiaries, which is intuitive. **BMI** also shows a positive correlation (0.20) with charges, indicating higher charges for beneficiaries with higher BMI.

The number of **children** has a very small positive correlation with charges (0.07), which is relatively insignificant.

Smoker(dummy) has a strong correlation to charges.

Region and **sex (dummy)** are weakly correlated to charges.

Regression Model and Interpretation

```
Model 2: OLS, using observations 1-1338
Dependent variable: charges

                  coefficient   std. error   t-ratio     p-value
  ---------------------------------------------------------------
  const           -13205.5      1118.29      -11.81      1.13e-030 ***
  age               256.968       11.8977     21.60      6.22e-089 ***
  bmi               337.025       28.3968     11.87      5.91e-031 ***
  children          468.354      137.745       3.400     0.0007    ***
  smoker_dummy    23867.1        412.691      57.83      0.0000    ***
  region_dummy      297.837      152.286       1.956     0.0507    *
  sex_dummy        -128.956      333.008      -0.3872    0.6986

Mean dependent var   13270.42   S.D. dependent var   12110.01
Sum squared resid    4.89e+10   S.E. of regression   6063.299
R-squared            0.750440   Adjusted R-squared   0.749315
F(6, 1331)           667.0634   P-value(F)           0.000000
Log-likelihood      -13549.02   Akaike criterion     27112.05
Schwarz criterion    27148.44   Hannan-Quinn         27125.68

Excluding the constant, p-value was highest for variable 3 (sex_dummy)
```

From this initial regression, we found out that variables ‘sex_dummy’ are statistically insignificant, so we drop them in the next regression.

```
Model 1: OLS, using observations 1-1338
Dependent variable: charges

                  coefficient   std. error   t-ratio    p-value
  ---------------------------------------------------------------
  const           -12102.8       941.984     -12.85    1.05e-035 ***
  age                257.850      11.8964     21.67    1.75e-089 ***
  bmi                321.851      27.3776     11.76    1.97e-030 ***
  children           473.502     137.792       3.436   0.0006    ***
  smoker_dummy     23811.4       411.220      57.90    0.0000    ***

Mean dependent var   13270.42   S.D. dependent var   12110.01
Sum squared resid    4.91e+10   S.E. of regression   6067.787
R-squared            0.749695   Adjusted R-squared   0.748943
F(4, 1333)           998.1232   P-value(F)           0.000000
Log-likelihood      -13551.02   Akaike criterion     27112.04
Schwarz criterion    27138.03   Hannan-Quinn         27121.77
```

In this regression, we see that there is no significant drop in R-squared by dropping the 'sex_dummy' and 'region_dummy' fields.

1. **Age**: The coefficient for age is 257.850, indicating that, on average, insurance charges increase by $257.850 for each additional year of age. This relationship is statistically significant ($p < 0.001$), suggesting that age is a significant predictor of insurance charges.
2. **BMI (Body Mass Index)**: The coefficient for BMI is 321.851, suggesting that, on average, insurance charges increase by $321.851 for each unit increase in BMI. This relationship is statistically significant ($p < 0.001$), indicating that BMI is a significant predictor of insurance charges.
3. **Children**: The coefficient for the number of children is 473.502, implying that, on average, insurance charges increase by $473.502 for each additional child. This relationship is statistically significant ($p < 0.001$), suggesting that the number of children is a significant predictor of insurance charges.
4. **Smoker (Smoking Status)**: The coefficient for the smoker dummy variable is 23811.4, indicating that smokers pay $23811.4 more, on average, for insurance compared to non-smokers. This relationship is highly statistically significant ($p < 0.001$), highlighting smoking status as a significant predictor of insurance charges.
5. **R-squared**: The R-squared value of 0.749695 indicates that approximately 74.97% of the variance in insurance charges can be explained by the independent variables (age, BMI, number of children, and smoking status) included in the model. This suggests that the model provides a good fit to the data, explaining a substantial portion of the variability in insurance charges

Test for assumption violation

Multicollinearity

gretl: collinearity

```
Variance Inflation Factors
Minimum possible value = 1.0
Values > 10.0 may indicate a collinearity problem

region_dummy     1.079
smoker_dummy     1.004
    children     1.002
         bmi     1.088
         age     1.016

VIF(j) = 1/(1 - R(j)^2), where R(j) is the multiple correlation coefficient
between variable j and the other independent variables

Belsley-Kuh-Welsch collinearity diagnostics:

  variance proportions

    lambda      cond     const region_d~ smoker_d~  children       bmi       age
     4.497     1.000     0.001     0.007     0.011     0.015     0.002     0.005
     0.768     2.420     0.000     0.003     0.956     0.023     0.000     0.002
     0.482     3.055     0.001     0.017     0.013     0.952     0.002     0.009
     0.161     5.279     0.002     0.713     0.006     0.004     0.022     0.118
     0.077     7.628     0.031     0.027     0.008     0.003     0.138     0.834
     0.014    17.612     0.965     0.232     0.006     0.003     0.836     0.033

  lambda = eigenvalues of inverse covariance matrix (smallest is 0.0144975)
  cond   = condition index
  note: variance proportions columns sum to 1.0

According to BKW, cond >= 30 indicates "strong" near linear dependence,
and cond between 10 and 30 "moderately strong".  Parameter estimates whose
variance is mostly associated with problematic cond values may themselves
be considered problematic.

Count of condition indices >= 30: 0

Count of condition indices >= 10: 1
Variance proportions >= 0.5 associated with cond >= 10:

   const     bmi
   0.965   0.836
```

- VIF values close to 1 indicate low multicollinearity, which is desirable. None of the variables have VIF values exceeding 10, suggesting no severe collinearity issues.
- None of the condition indices exceed 30, indicating no strong near-linear dependence among the independent variables.
- One condition index exceeds 10, suggesting moderately strong multicollinearity.

Assumption is not violated.

Heteroscedasticity

```
modtest --white

ite's test for heteroskedasticity
S, using observations 1-1338
pendent variable: uhat^2

                     coefficient        std. error     t-ratio   p-value
-----------------------------------------------------------------------
const                  1.52948e+08       7.21122e+07   2.121     0.0341  **
age                  285277              1.32832e+06   0.2148    0.8300
bmi                   -8.10654e+06       3.38221e+06  -2.397     0.0167  **
children               7.89556e+06       1.28179e+07   0.6160    0.5380
smoker_dummy           7.24459e+07       3.39653e+07   2.133     0.0331  **
region_dummy          -2.02631e+07       1.74237e+07  -1.163     0.2451
sq_age                 6866.35          12946.4        0.5304    0.5959
X2_X3                 -8189.30          25463.1       -0.3216    0.7478
X2_X4                 -66653.6         132487         -0.5031    0.6150
X2_X5                 225686           369901          0.6101    0.5419
X2_X6                -237932           136111         -1.748     0.0807  *
sq_bmi                118898            45612.2        2.607     0.0092  ***
X3_X4                  22324.9         296002          0.07542   0.9399
X3_X5                -712631           858971         -0.8296    0.4069
X3_X6                 590371           352940          1.673     0.0946  *
sq_children           -2.44215e+06       1.28556e+06  -1.900     0.0577  *
X4_X5                 -6.83279e+06       4.44941e+06  -1.536     0.1249
X4_X6                  1.98499e+06       1.60463e+06   1.237     0.2163
X5_X6                  2.11969e+06       4.68586e+06   0.4524    0.6511
sq_region_dummy        2.06917e+06       2.13498e+06   0.9692    0.3326

Unadjusted R-squared = 0.101367

st statistic: TR^2 = 135.628940,
th p-value = P(Chi-square(19) > 135.628940) = 0.000000
```

- The test statistic is 128.470656.
- The p-value associated with the test statistic is extremely low ($p < 0.0001$).
- Indicating significant evidence to not accept the null hypothesis of homoscedasticity, suggesting the presence of heteroskedasticity in the residuals.
- Implies that the variance of the errors is not constant across all levels of the independent variables.

Assumption is violated.

<u>This can be solved by running a WLS with the inverse of squared residuals as weights</u>

Autocorrelation

```
Using numerical derivatives
Tolerance = 1.81899e-012
Convergence achieved after 40 iterations

Model 8: NLS, using observations 2-1338 (n = 1337)
l_charges = beta1*(1-rho) + rho*l_charges(-1)+ beta2*(l_age-rho*l_age(-1))+beta3*(l_bmi-rho*l_bmi(-1))+beta5*(l_smoker-rho*l_smoker(-1))+beta6*(l_region_dummy-rho*l_region_dummy(-1))

             estimate    std. error   t-ratio     p-value
  ---------------------------------------------------------
  beta1       4.24446     0.241793      17.55     3.26e-062 ***
  beta2       1.27090     0.0321963     39.47     2.96e-226 ***
  beta3       0.423151    0.06443345     6.567    7.33e-011 ***
  beta5      -2.23525     0.0447580    -49.94     2.11e-307 ***
  beta6       0.0965095   0.02444648     3.945    8.40e-05  ***
  rho        -0.0167356   0.0274599     -0.6095   0.5423

Mean dependent var   9.098183   S.D. dependent var   0.919707
Sum squared resid    277.2760   S.E. of regression   0.456423
R-squared            0.754638   Adjusted R-squared   0.753716
Log-likelihood      -845.4564   Akaike criterion     1702.913
Schwarz criterion    1734.102   Hannan-Quinn         1714.599

GNR: R-squared = 6.72795e-014, max |t| = 9.32156e-006
Convergence seems to be reasonably complete
```

Conclusion

Our thorough analysis of insurance pricing factors revealed nuanced insights into premium determinants. Age emerged as a significant predictor, indicating higher premiums for older individuals, with a notable spike at age 18, possibly reflecting targeted marketing strategies.

BMI strongly influenced premiums, with higher charges for individuals with elevated BMI values, highlighting the impact of lifestyle on insurance costs. Similarly, the number of children correlated with higher premiums, reflecting the financial implications of family size on healthcare expenses.

Smoking status had a profound effect, with smokers facing substantially higher premiums. This underscores the increased health risks and costs associated with tobacco use, necessitating differentiated pricing strategies.

While our study provided valuable insights, it also uncovered violations of assumptions like heteroskedasticity. Addressing these limitations is crucial for refining predictive models and enhancing pricing accuracy.

In conclusion, our analysis contributes to a deeper understanding of insurance dynamics, emphasizing the importance of diverse factors in pricing strategies. By leveraging advanced

analytics, insurers can develop more accurate and equitable pricing models, benefiting both insurers and policyholders.